INTERNATIONAL MONETARY FUND

HAITI

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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The Debt Sustainability Analysis (DSA) was prepared in accordance with the revised joint Bank-Fund debt sustainability framework (DSF) for low-income countries (LICs).[1] Haiti's risk of debt distress has increased from moderate to high[2], due to a decline in long-term growth prospects, an increase in projected borrowing, and the impact of devastating Hurricane Matthew. Reconstruction following the hurricane is expected to require large increases in public spending, with budget revenues remaining broadly flat. While aid inflows may lead to some real appreciation and improvement in debt ratios over the short term, the real exchange rate will eventually depreciate towards its long-run equilibrium, contributing to an eventual increase in debt ratios. Similarly, real output growth is expected to return eventually to its historical average annual pace of 2 percent. This baseline scenario assumes that improvements in energy policies will allow for fiscal gains from lower global oil prices, and that there will be a gradual rise in public investment spending outside of post-hurricane reconstruction efforts.

There are risks to this baseline scenario. Haiti's debt profile is vulnerable to fluctuations in exchange rates, external financing terms, output growth, and oil prices. A customized scenario assuming weak output performance (accompanied by lower export growth and reduced FDI flows) is used to inform the external risk rating. In order to improve its debt profile, lock in advances in public financial management, and deepen growth-enhancing structural reforms, Haiti needs to maintain prudent macroeconomic policies, avoid contracting debt on nonconcessional terms, and move towards greater political stability.

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[1] IMF and World Bank (2012), "Revisiting the Debt Sustainability Framework for Low-Income Countries," (SM/12/10). In line with the guidance note, "a change in the external risk rating or in the assessment of the overall risk of debt distress since the previous DSA would warrant a full DSA". Accordingly, this full DSA was prepared jointly by IMF and World Bank staff. Previously, a full DSA prepared in May 2015 classified Haiti's risk of debt distress as "moderate".

[2] Haiti is classified as "weak" policy performer, based on the quality of its policies and institutions as measured by a three-year average of the World Bank's Country Policy and Institutional Assessment (CPIA) Index. Haiti's average CPIA score (2013–15) is 2.86. The relevant indicative thresholds for "weak performers" for external debt sustainability are PV of debt-to-GDP ratio of 30 percent and 27 percent when remittances are included; PV of debt-to-exports ratio of 100 percent and 80 percent when remittances are included, PV of debt-to-revenue ratio of 200 percent, debt service-to-exports ratio of 15 percent and 12 percent with remittances; and debt service-to-revenue ratio of 25 percent.

BACKGROUND

1. At the end of 2016, Haiti's stock of public sector debt totaled US$2.4 billion (28.8 percent of GDP), composed mainly of external debt on concessional terms (Text Table 1). External public debt accounts for the majority of total outstanding public debt (24 percent of GDP), about three-quarters of which is debt stemming from oil imports financed by Venezuela's Petrocaribe, generally on concessional terms. The remainder is largely concessional debt from multilateral institutions. Domestic public debt amounts to US$112 million, mostly in the form of medium term treasury bills held by commercial banks. External arrears to Venezuela accumulated in FY 2015/16 for unpaid fuel deliveries from Petrocaribe (approximately US$47 million) are in the process of being restructured.

Text Table 1. Haiti: Structure of External Public Debt at end-2016
(Fiscal-year basis)

		in percent of	
	US$ millions	total debt	GDP 1/
Total	**2038.6**	**100.0**	**24.7**
Multilateral creditors	**163.2**	**8.0**	**2.0**
o/w IMF	65.3	3.2	0.8
o/w OPEC	43.9	2.2	0.5
o/w IFAD	54.0	2.6	0.7
o/w IDA	0.0	0.0	0.0
Bilateral creditors	**1875.4**	**92.0**	**22.7**
Venezuela	1790.4	87.8	21.7
o/w PetroCaribe	1763.7	86.5	21.4
o/w BANDES	26.7	1.3	0.3
Taiwan, Province of China	84.9	4.2	1.0
Other borrowing	**0.0**	**0.0**	**0.0**

Sources: Haitian authorities, and Fund staff estimates.

1/ The debt ratio differs slightly from that in Table 3 given the use of average, instead of end-of-period, exchange rates.

2. In October 2016, Haiti was hit by Hurricane Matthew. The damage caused by the hurricane will have a significant impact on real output growth and debt burden indicators. The Rapid Damage and Loss Assessment (RDLA) report estimated the total damage and losses to the economy caused by Hurricane Matthew at US$1.9 billion, or about 23 percent of GDP, of which the damage to physical assets accounts for 15 percent of GDP, while production losses account for 7 percent of GDP – mostly due to losses in agriculture.

3. Post-hurricane reconstruction needs are substantial and will require external assistance as well as a protracted increase in fiscal deficits. Although some of the external assistance will be provided on grant terms and part of the reconstruction costs will be borne by the private sector, the overall fiscal deficit is expected to be higher compared to the trajectory projected in the previous DSA (Text Table 2).

MACROECONOMIC ASSUMPTIONS, 2016–2036

4. Haiti's fiscal consolidation is expected to be reversed in the aftermath of Hurricane Matthew. The fiscal consolidation effort reduced the primary deficit from an average of 6.4 percent of GDP in 2013–2014 to 1.7 percent of GDP in 2015–2016. The adjustment was mostly based on a reduction in primary capital spending. As a result of the hurricane, a previously anticipated rebound in growth to an average of 3.7 percent over the next four years has been adjusted downward to an average below 3 percent, and the country is expected to move down to a lower growth path in the long term. Consumer prices are expected to move up in the short term relative to previous forecasts, forced up by increases in domestic food prices arising from the destruction of productive capacity.

5. Specifically, the baseline scenario is based on the following assumptions (Text Table 2):

- Real GDP is expected to grow by 2.5 percent in 2017, accelerating to 3 percent from 2018 to 2021 as investment associated with reconstruction efforts raises economic output and agricultural production recovers. Over the long run, the average annual growth rate is expected to return to 2 percent in the absence of deeper structural reforms that would allow for more robust long-term growth.

- A stable real exchange rate vis-à-vis the US dollar is expected over the longer term, after some initial movement caused by the rise in prices of domestically produced goods. Over the medium to longer term, inflation is expected to stabilize at 5 percent.

- Oil prices are projected to decline in real terms in line with WEO forecasts during 2016-20 and to remain constant thereafter.

- Although the positive fiscal impact of lower oil prices is locked in, damage caused by Hurricane Matthew will reverse the fiscal gains achieved in 2014 and 2015. Primary spending (including EDH's deficit) is set to pick up from 18 percent of GDP in 2016 to close to 23 percent of GDP in 2017–20, while revenues excluding grants are assumed to remain on the order of 14 to 15 percent of GDP over the medium term.

- The steady improvement in the external balance of goods and services that has been underway since 2010 is expected to reverse, due to reconstruction activity. Subsequently, the trade deficit is expected to gradually narrow through the projection period, while the current account deficit converges to about 2 percent of GDP.

Text Table 2. Haiti: 2015 DSA vs. 2016 DSA

	2016		2017-20 Avg		2021-35 Avg	
	Previous DSA	Current DSA	Previous DSA	Current DSA	Previous DSA	Current DSA
(annual percentage change, unless otherwise indicated)						
Real GDP	3.2	1.5	3.7	2.9	3.3	2.1
Consumer prices (period average)	6.5	13.3	5.1	7.0	5.0	5.0
(in percent of GDP, unless otherwise indicated)						
Total revenue and grants	21.9	17.0	22.5	20.0	20.5	19.1
Of which: Revenue	16.2	13.9	17.5	14.6	17.1	17.4
Primary expenditure	26.4	18.0	25.7	22.9	23.5	21.5
Of which: Capital expenditure	14.1	6.4	13.1	11.9	10.9	10.0
Overall balance	-2.3	-1.6	-2.1	-3.4	-1.9	-3.0
Current account balance	-3.8	0.4	-3.6	-2.6	-3.4	-1.9
Exports of goods and services	19.3	21.2	19.8	22.7	22.4	26.5
Imports of goods and services	-47.1	-49.5	-46.6	-56.7	-41.8	-52.1

Sources: Haitian authorities; and Fund staff estimates and projections.

6. After falling by half in 2015, the fiscal gross financing need is expected to rise sharply over the medium term. The fiscal gross financing need dropped from 4.5 percent of GDP in 2015 to 3.0 percent in 2016. Following Hurricane Matthew, the gross financing need is set to increase to about 6 percent of GDP in 2017 rising to almost 13 percent of GDP by 2031 as public debt accumulates and declining thereafter as reliance on concessional borrowing is gradually replaced by FDI inflows.

DEBT SUSTAINABILITY OUTLOOK, 2016–2036

7. The previous DSA cautioned against Haiti's vulnerabilities even as it assigned a moderate level of distress risk. Although providing a more favorable assessment than the present DSA, the 2015 DSA underscored that the balance of risks remained to the downside. Some of these risks have materialized since the last assessment. In 2015 and 2016, growth has slowed down against a background of a severe drought affecting the agricultural sector, and lower investments stemming from public financing constraints amid continued political uncertainty. The exchange rate also depreciated faster than expected, and consumer price inflation accelerated relative to expectations. Finally, the country has suffered major losses in the aftermath of Hurricane Matthew. Reconstruction efforts should allow growth to pick up in the short term and would also improve access to concessional financing. Those factors have weighed on the country's outlook and external debt assessment.

8. After graduating from high to moderate risk rating in 2015, shocks sustained due to Hurricane Matthew shift Haiti back to high risk of external debt distress. Given the importance of remittances in enhancing Haiti's capacity to repay its external debt, remittances are included in external debt thresholds in this DSA (Figure 1). Under the baseline scenario with added remittances, there is one breach, and two near-breaches of the policy-related indicative thresholds. The breach occurs to the present value (PV) of external debt divided by the sum of GDP and remittances from 2029 until the end of the projection period. Furthermore, by 2028, the PV of external debt is projected to rise above 32 percent of GDP, exceeding the upper band for thresholds determining borderline cases (2013 Guidance Note, Table 7 on p. 39), indicating high risk of debt distress.[3] When remittances are excluded from the denominator as a sensitivity check, debt indicators also breach the threshold for PV of debt-to-exports ratio from 2020 onwards (Figure 2).

9. Public sector debt indicators point to an overall high risk of debt distress. Under the baseline scenario, the PV of total public debt-to-GDP ratio breaches the indicative threshold of 38 percent from 2027 onward (Figure 3). This ratio rises to peak at 43 percent in 2032–33 and declines thereafter until the end of the projection period. Two factors drive the increase in this ratio over the next ten to fifteen years.

[3] While Haiti's high risk of debt distress is well-defined by the traditional approach of the DSA, the assessment through the probability approach also confirms this result. Under the probability approach, Haiti is clearly at high risk of debt distress, with all three solvency debt burden indicators continuously above the indicative threshold. The probability approach differs from the traditional approach by providing indicative thresholds based on country-specific CPIAs rather than performance groups (weak, medium, strong performer). In light of Haiti's relatively low policy and institutional capacity within the weak performance group, all three solvency as well as the liquidity indicators breach the indicative thresholds. While the solvency indicators remain well above the indicative thresholds throughout the projection period, the liquidity indicators breach the thresholds toward the end of the projection period.

The first is the reversion of the GDP growth rate down to an annual average rate of 2 percent, closer to, although still much better than, the 30-year average growth rate of 0.8 percent. The second factor working to increase the ratio is a continued reliance on debt-creating (although nonetheless concessional) flows to fund the current account deficit. The debt-service-to-revenue ratio also increases under the baseline scenario, rising from 11 percent in 2016 to peak at 18 percent in 2035. After that peak, it begins to shift downward, tracking the trajectory of the public-debt-to-GDP ratio, but with a lag of a few years.

10. In the long run, debt sustainability indicators are expected to improve gradually. Absent major shocks, deterioration in macroeconomic policies, or extra borrowing on nonconcessional terms, the trajectory of Haiti's public and external debt sustainability indicators is expected to improve towards the end of the projection period. As Haiti develops its export capacity, including through the planned expansion of its light-manufacturing Free Trade Zones (FTZs), foreign direct investment inflows will eventually rise commensurate with a decline in external borrowing, improving the composition of external financing and benefiting Haiti's debt ratios in the longer term. The number of FTZs is projected to increase gradually over time, as the political situation stabilizes, and the damage from the hurricane and the 2010 earthquake is fully replaced.

11. Increased-stress scenarios paint a more pessimistic picture. As Table 2 indicates, public debt is most sensitive to real growth assumptions. The low-growth alternative scenario assumes real output growth at only 0.5 percent in FY 2016/17 and 1 percent thereafter[4], accompanied by weak import and export growth, and anemic FDI inflows stabilizing at 2 percent of GDP in the long run. Under this scenario, the debt distress threshold of 30 percent for PV of public debt-to-GDP ratio is breached one year sooner than in the baseline, and the projected path of various debt indicators does not improve throughout the projection period (Figure 2). Under the historical scenario which preserves key macroeconomic indicators at their ten-year average level, the PV of debt-to-GDP + remittances ratio breaches its indicative threshold sooner than under the baseline scenario, while its path does not improve over the projection horizon. In the historical scenario, the PV of debt-to-exports + remittances ratio breaches its distress threshold in 2029, which does not happen using baseline assumptions (Figure 1). These increased-stress scenarios underscore Haiti's vulnerability to policy shocks, which can send its debt burden on an upward trend. Furthermore, they show that the debt profile remains sensitive to non-debt creating flows, in this case, the assumptions on FDI in the financing mix.

CONCLUSION AND AUTHORITIES' VIEWS

12. The updated DSA indicates that Haiti's risk of debt distress has shifted from moderate to high. Reconstruction efforts and production losses following Hurricane Matthew are expected to set public spending on an upward trajectory, with budget revenues remaining broadly flat. In addition, real output growth is likely to slow and possibly place Haiti on a permanently lower growth path compared to the outlook shown in the last DSA. With little domestic financing available, the rising gross financing need will be satisfied with external grants and concessional borrowing. The latter will send debt burden indicators on

[4] Real output growth averaged 0.8 percent annually between 1986 and 2016.

an upward trajectory, which under the traditional approach results in the breach of one indicative threshold (PV of debt-to-GDP ratio).

13. The authorities broadly concurred with the main findings of the DSA. Staff and the authorities discussed the main assumptions and conclusions of the updated DSA, and the authorities broadly agreed with the risk of debt distress assessment.

Table 1. Haiti: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–36 1/

(In percent of GDP, unless otherwise indicated; fiscal-year basis)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	19.0	24.1	26.2			28.8	28.9	32.7	35.3	37.2	37.7		48.6	53.9	
of which: foreign-currency denominated	17.7	21.2	24.3			27.4	27.0	30.4	32.8	34.6	34.9		42.2	45.3	
Change in public sector debt	5.2	5.1	2.1			2.6	0.1	3.8	2.5	1.9	0.6		2.4	-1.0	
Identified debt-creating flows	6.1	6.3	1.4			3.6	1.7	3.3	2.2	1.7	0.4		1.9	-2.4	
Primary deficit	6.9	6.2	2.2	3.4	1.9	1.2	4.1	3.0	2.3	2.1	1.6	2.4	3.1	1.0	2.2
Revenue and grants	20.9	18.9	19.3			17.0	20.6	20.0	19.9	19.7	19.8		19.1	19.7	
of which: grants	8.1	6.4	5.4			3.1	6.8	5.5	5.0	4.6	4.2		2.1	0.0	
Primary (noninterest) expenditure	27.8	25.0	21.4			18.2	24.7	22.9	22.1	21.8	21.3		22.2	20.7	
Automatic debt dynamics	-0.8	-0.5	1.2			2.6	-2.3	0.4	-0.1	-0.5	-1.2		-1.2	-1.8	
Contribution from interest rate/growth differential	-0.8	-0.6	0.2			1.5	0.2	-0.4	0.1	-0.3	-0.8		-1.2	-1.8	
of which: contribution from average real interest rate	-0.2	-0.1	0.5			1.9	0.9	0.5	1.1	0.7	0.2		-0.3	-0.7	
of which: contribution from real GDP growth	-0.6	-0.5	-0.3			-0.4	-0.7	-0.8	-1.0	-1.0	-1.1		-0.9	-1.1	
Contribution from real exchange rate depreciation	0.0	0.1	1.0			1.1	-2.5	0.8	-0.3	-0.1	-0.3		...	...	
Other identified debt-creating flows	0.0	0.6	-1.9			-0.1	-0.1	0.0	0.0	0.0	0.0		0.0	-1.6	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	-1.6	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.6	-1.9			-0.1	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.9	-1.1	0.7			-1.1	-1.5	0.5	0.4	0.2	0.2		0.5	1.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	18.4			20.3	20.6	23.4	25.1	26.5	27.1		36.2	41.7	
of which: foreign-currency denominated	...	...	16.4			19.0	18.8	21.0	22.7	24.0	24.3		29.9	33.2	
of which: external	...	...	16.4			19.0	18.8	21.0	22.7	24.0	24.3		29.9	33.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.8	8.3	4.5			3.0	6.1	5.9	6.1	6.1	5.8		11.0	11.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	95.3			119.7	100.0	117.0	126.6	134.5	137.2		190.1	211.3	
PV of public sector debt-to-revenue ratio (in percent)	...	...	132.8			146.4	148.9	162.0	169.0	175.8	174.5		214.1	211.3	
of which: external 3/	...	...	118.8			136.7	135.6	145.9	152.6	158.8	156.2		176.4	168.0	
Debt service-to-revenue and grants ratio (in percent) 4/	4.2	5.0	5.6			9.2	8.9	9.3	10.0	9.2	9.3		12.9	14.8	
Debt service-to-revenue ratio (in percent) 4/	6.9	7.5	7.9			11.3	13.2	12.9	13.3	12.0	11.9		14.5	14.8	
Primary deficit that stabilizes the debt-to-GDP ratio	1.7	1.0	0.1			-1.4	3.9	-0.9	-0.3	0.2	1.0		0.7	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	4.2	2.8	1.2	2.1	3.0	1.5	2.5	3.0	3.0	3.0	3.0	2.7	2.0	2.0	2.0
Average nominal interest rate on forex debt (in percent)	1.2	1.0	0.8	1.0	0.4	0.9	1.2	1.3	1.3	1.4	1.4	1.2	1.3	1.2	1.3
Average real interest rate on domestic debt (in percent)	...	-2.5	-4.6	-2.8	1.7	-1.6	3.7	7.0	6.7	4.9	4.5	4.2	-1.6	...	-1.3
Real exchange rate depreciation (in percent, + indicates depreciation	-0.4	0.6	4.6	-0.3	4.3	4.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	6.6	4.7	7.2	7.6	3.6	12.1	10.0	5.0	5.0	5.0	5.0	7.0	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percer	1.7	-7.5	-13.4	-1.8	4.8	-14.0	39.4	-4.4	-0.5	1.6	0.7	3.8	1.6	4.8	1.8
Grant element of new external borrowing (in percent)	...	...	...	...	...	33.6	37.0	38.3	37.9	37.7	36.6	36.9	38.8	38.2	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Haiti: Sensitivity Analysis for Key Indicators of Public Debt, 2016–36

(Fiscal-year basis)

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	20	21	23	25	27	27	36	42
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	20	20	23	26	29	31	42	57
A2. Primary balance is unchanged from 2016	20	18	20	21	22	22	25	28
A3. Alternative Scenario: Low Growth 1/	20	21	24	26	28	29	40	52
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	20	22	26	29	31	33	47	61
B2. Primary balance is at historical average minus one standard deviations in 2017-201	20	22	26	28	29	30	39	44
B3. Combination of B1-B2 using one half standard deviation shocks	20	21	26	28	30	31	43	53
B4. One-time 30 percent real depreciation in 2017	20	28	30	32	33	33	41	49
B5. 10 percent of GDP increase in other debt-creating flows in 2017	20	29	31	32	34	34	43	47
PV of Debt-to-Revenue Ratio 2/								
Baseline	120	100	117	127	135	137	190	211
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	120	98	117	132	146	157	220	291
A2. Primary balance is unchanged from 2016	120	89	99	105	109	112	133	140
A3. Alternative Scenario: Low Growth 1/	120	100	117	128	136	141	201	265
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	120	104	129	144	157	164	244	311
B2. Primary balance is at historical average minus one standard deviations in 2017-201	120	105	131	140	148	150	203	221
B3. Combination of B1-B2 using one half standard deviation shocks	120	102	127	139	150	155	223	271
B4. One-time 30 percent real depreciation in 2017	120	136	152	160	167	168	217	247
B5. 10 percent of GDP increase in other debt-creating flows in 2017	120	139	156	163	171	172	225	239
Debt Service-to-Revenue Ratio 2/								
Baseline	9	9	9	10	9	9	13	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	9	9	9	10	11	16	24
A2. Primary balance is unchanged from 2016	9	9	9	6	5	5	7	7
A3. Alternative Scenario: Low Growth 1/	9	9	9	10	9	9	13	18
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	9	9	10	11	12	12	18	26
B2. Primary balance is at historical average minus one standard deviations in 2017-201	9	9	10	12	13	11	14	16
B3. Combination of B1-B2 using one half standard deviation shocks	9	9	10	11	11	11	16	21
B4. One-time 30 percent real depreciation in 2017	9	10	12	14	14	14	19	28
B5. 10 percent of GDP increase in other debt-creating flows in 2017	9	9	12	23	14	14	16	19

Sources: Country authorities; and staff estimates and projections.

1/ Assumes real annual GDP growth at 0.5 percent in 2017 and 1 percent thereafter.

2/ Revenues are defined inclusive of grants.

Table 3. Haiti: External Debt Sustainability Framework, Baseline Scenario, 2013–36 1/

(In percent of GDP, unless otherwise indicated; fiscal-year basis)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-20 Averag
External debt (nominal) 1/	**17.7**	**21.2**	**24.3**			**27.4**	**27.0**	**30.4**	**32.8**	**34.6**	**34.9**		**42.2**	**45.3**	
of which: public and publicly guaranteed (PPG)	17.7	21.2	24.3			27.4	27.0	30.4	32.8	34.6	34.9		42.2	45.3	
Change in external debt	3.9	3.5	3.0			3.2	-0.4	3.4	2.4	1.8	0.3		1.9	-1.0	
Identified net debt-creating flows	3.5	4.5	1.5			-1.5	1.5	0.6	0.4	-0.4	-0.9		0.2	-2.5	
Non-interest current account deficit	**6.2**	**6.1**	**2.3**	**3.3**	**2.1**	**-0.6**	**3.6**	**2.1**	**2.1**	**1.4**	**1.1**		**1.5**	**1.6**	
Deficit in balance of goods and services	33.7	32.5	30.6			28.2	36.8	34.4	33.3	31.9	30.6		26.9	20.7	
Exports	18.6	18.9	20.2			21.2	21.7	22.2	23.1	23.7	24.0		25.7	29.7	
Imports	52.3	51.4	50.8			49.5	58.4	56.5	56.4	55.6	54.7		52.6	50.4	
Net current transfers (negative = inflow)	-27.0	-26.1	-28.0	-30.1	7.0	-28.8	-33.1	-32.1	-30.9	-30.1	-29.2		-24.7	-18.6	-2
of which: official	-8.9	-6.4	-5.6			-3.3	-7.1	-5.5	-5.0	-4.6	-4.2		-2.1	0.0	
Other current account flows (negative = net inflow)	-0.5	-0.3	-0.3			-0.1	-0.1	-0.2	-0.3	-0.4	-0.3		-0.6	-0.5	
Net FDI (negative = inflow)	**-1.9**	**-1.1**	**-1.2**	**-1.6**	**0.9**	**-0.7**	**-1.7**	**-1.0**	**-1.2**	**-1.3**	**-1.5**		**-1.1**	**-3.7**	
Endogenous debt dynamics 2/	**-0.8**	**-0.5**	**0.4**			**-0.2**	**-0.4**	**-0.5**	**-0.5**	**-0.5**	**-0.5**		**-0.3**	**-0.4**	
Contribution from nominal interest rate	0.2	0.2	0.2			0.2	0.3	0.3	0.4	0.4	0.5		0.5	0.5	
Contribution from real GDP growth	-0.5	-0.5	-0.3			-0.4	-0.7	-0.8	-0.9	-1.0	-1.0		-0.8	-0.9	
Contribution from price and exchange rate changes	-0.4	-0.2	0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**0.4**	**-0.9**	**1.5**			**4.7**	**-1.9**	**2.7**	**2.0**	**2.2**	**1.2**		**1.7**	**1.5**	
of which: exceptional financing	-0.1	-0.4	-0.4			0.0	-0.3	-0.3	-0.1	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	16.4			19.0	18.8	21.0	22.7	24.0	24.3		29.9	33.2	
In percent of exports	...	...	81.2			89.4	86.7	94.9	98.4	101.0	101.1		116.2	111.6	
PV of PPG external debt	**...**	**...**	**16.4**			**19.0**	**18.8**	**21.0**	**22.7**	**24.0**	**24.3**		**29.9**	**33.2**	
In percent of exports	**...**	**...**	**81.2**			**89.4**	**86.7**	**94.9**	**98.4**	**101.0**	**101.1**		**116.2**	**111.6**	
In percent of government revenues	**...**	**...**	**118.8**			**136.7**	**135.6**	**145.9**	**152.6**	**158.8**	**156.2**		**176.4**	**168.0**	
Debt service-to-exports ratio (in percent)	**1.8**	**2.4**	**2.5**			**4.2**	**5.9**	**6.2**	**6.4**	**6.7**	**6.7**		**6.5**	**9.8**	
PPG debt service-to-exports ratio (in percent)	**1.8**	**2.4**	**2.5**			**4.2**	**5.9**	**6.2**	**6.4**	**6.7**	**6.7**		**6.5**	**9.8**	
PPG debt service-to-revenue ratio (in percent)	**2.6**	**3.7**	**3.6**			**6.5**	**9.2**	**9.6**	**10.0**	**10.5**	**10.4**		**9.8**	**14.8**	
Total gross financing need (Billions of U.S. dollars)	0.4	0.5	0.1			0.0	0.3	0.2	0.2	0.1	0.1		0.2	0.1	
Non-interest current account deficit that stabilizes debt ratio	2.2	2.6	-0.7			-3.8	4.0	-1.3	-0.3	-0.4	0.8		-0.3	2.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	4.2	2.8	1.2	2.1	3.0	1.5	2.5	3.0	3.0	3.0	3.0	2.7	2.0	2.0	
GDP deflator in US dollar terms (change in percent)	2.8	1.2	-2.0	5.3	6.6	-6.6	-2.0	0.0	-1.8	-0.5	1.0	-1.7	1.9	1.9	
Effective interest rate (percent) 5/	1.2	1.0	0.8	1.0	0.4	0.9	1.2	1.3	1.3	1.4	1.4	1.2	1.3	1.2	
Growth of exports of G&S (US dollar terms, in percent)	18.5	6.0	6.0	11.6	9.2	-0.6	2.6	5.2	5.3	5.4	5.4	3.9	5.5	5.5	
Growth of imports of G&S (US dollar terms, in percent)	5.3	2.2	-2.0	10.2	17.0	-7.8	18.7	-0.4	0.8	1.2	2.3	2.5	3.4	4.1	
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	33.6	37.0	38.3	37.9	37.7	36.6	36.9	38.8	38.2	3
Government revenues (excluding grants, in percent of GDP)	12.8	12.5	13.8			13.9	13.9	14.4	14.9	15.1	15.6		16.9	19.7	1
Aid flows (in Billions of US dollars) 7/	18.4	17.6	9.5			0.3	0.7	0.7	0.6	0.6	0.5		0.6	0.3	
of which: Grants	0.7	0.6	0.5			0.3	0.6	0.5	0.4	0.4	0.4		0.2	0.0	
of which: Concessional loans	17.7	17.1	9.0			0.0	0.1	0.2	0.2	0.2	0.1		0.3	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.6	8.0	7.2	6.5	6.1	5.4		3.9	1.2	
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			79.4	79.8	72.9	72.2	71.8	73.3		58.5	38.2	5
Memorandum items:															
Nominal GDP (Billions of US dollars)	8.5	8.8	8.7			8.3	8.3	8.5	8.6	8.9	9.2		11.3	16.6	
Nominal dollar GDP growth	7.1	4.0	-0.9			-5.2	0.5	3.0	1.1	2.5	4.0	1.0	4.0	4.0	
PV of PPG external debt (in Billions of US dollars)			1.3			1.4	1.5	1.7	1.9	2.1	2.2		3.3	5.4	
(PVt-PVt-1)/GDPt-1 (in percent)						0.8	1.6	2.3	2.0	1.9	1.5	1.7	2.5	0.6	
Gross workers' remittances (Billions of US dollars)	1.5	1.7	1.9			2.1	2.2	2.3	2.2	2.3	2.3		2.5	3.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.4			15.1	14.9	16.6	18.0	19.1	19.4		24.4	28.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	38.6			40.6	39.4	43.2	46.4	48.7	49.6		61.8	68.6	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	1.2			1.9	2.7	2.8	3.0	3.2	3.3		3.4	6.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Haiti: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36

(In percent; fiscal-year basis)

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	19	19	21	23	24	24	**30**	33
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	19	17	19	20	22	23	**29**	38
A2. New public sector loans on less favorable terms in 2016-2036 2	19	19	22	25	27	28	**38**	48
A3. Alternative Scenario : Low growth	17	17	20	22	24	25	**31**	41
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	19	19	22	24	25	26	**32**	35
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	19	19	21	22	24	24	**30**	33
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	19	18	20	22	23	24	**30**	33
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	19	25	33	35	37	37	**40**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	19	22	25	27	29	29	**34**	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	19	27	29	32	34	34	**42**	47
PV of debt-to-exports ratio								
Baseline	89	87	95	98	101	101	**116**	112
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	89	80	87	87	91	94	**114**	126
A2. New public sector loans on less favorable terms in 2016-2036 2	89	89	99	107	113	117	**146**	163
A3. Alternative Scenario : Low growth	80	77	87	94	100	105	**135**	193
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	89	86	91	95	98	99	**114**	110
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	89	86	96	100	103	104	**119**	113
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	89	86	91	95	98	99	**114**	110
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	89	118	149	153	155	156	**157**	118
B5. Combination of B1-B4 using one-half standard deviation shocks	89	97	110	114	116	117	**125**	107
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	89	86	91	95	98	99	**114**	110
PV of debt-to-revenue ratio								
Baseline	137	136	146	153	159	156	**176**	168
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	137	125	133	135	142	145	**173**	190
A2. New public sector loans on less favorable terms in 2016-2036 2	137	139	152	165	178	180	**222**	245
A3. Alternative Scenario : Low growth	123	123	136	146	156	159	**182**	209
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	137	139	151	159	166	165	**187**	178
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	137	134	143	151	158	156	**176**	166
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	137	133	141	149	156	155	**174**	166
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	137	184	229	237	244	241	**239**	178
B5. Combination of B1-B4 using one-half standard deviation shocks	137	156	176	184	191	189	**199**	168
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	137	193	202	213	223	221	**250**	238

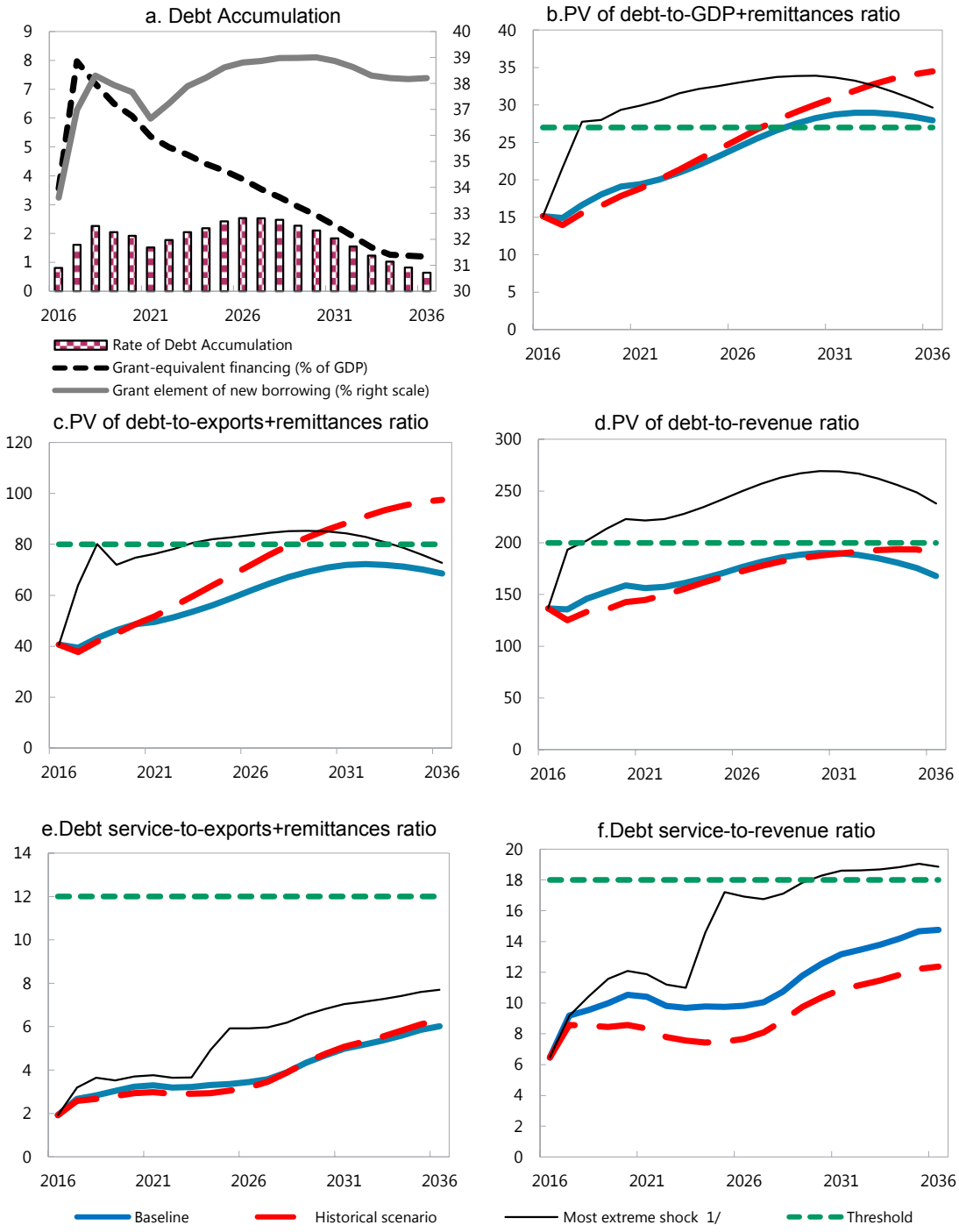
Table 4. Haiti: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36 (concluded)

Debt service-to-exports ratio								
Baseline	4	6	6	6	7	7	**6**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	4	5	6	5	5	5	**5**	8
A2. New public sector loans on less favorable terms in 2016-2036 2	4	6	6	7	7	7	**11**	16
A3. Alternative Scenario : Low growth	4	5	6	6	7	7	**8**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	4	6	6	6	7	7	**6**	10
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	4	6	6	7	7	7	**7**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	4	6	6	6	7	7	**6**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	4	6	7	7	8	8	**11**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	4	6	6	6	7	7	**8**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	4	6	6	6	7	7	**6**	10
Debt service-to-revenue ratio								
Baseline	6	9	10	10	11	10	**10**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	6	9	9	8	9	8	**8**	12
A2. New public sector loans on less favorable terms in 2016-2036 2	6	9	10	10	11	11	**16**	23
A3. Alternative Scenario : Low growth	6	9	9	10	11	11	**10**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	6	9	10	11	11	11	**11**	16
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	6	9	10	10	11	10	**10**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	6	9	10	10	11	10	**10**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	6	9	10	12	12	12	**17**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	6	9	10	10	11	11	**13**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	6	13	14	14	15	15	**14**	21
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	37	37	37	37	37	37	**37**	37

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

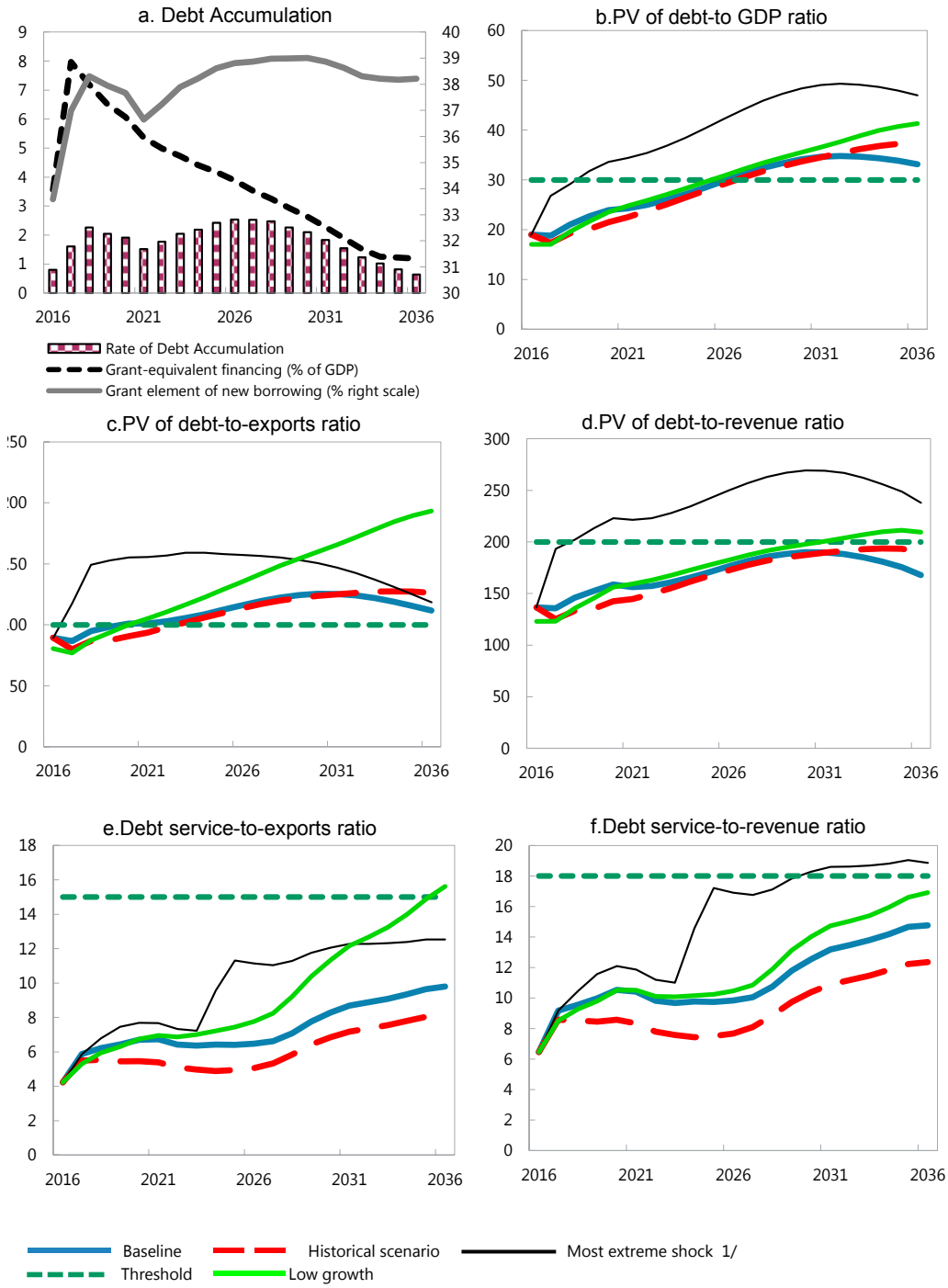
Figure 1. Haiti: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock

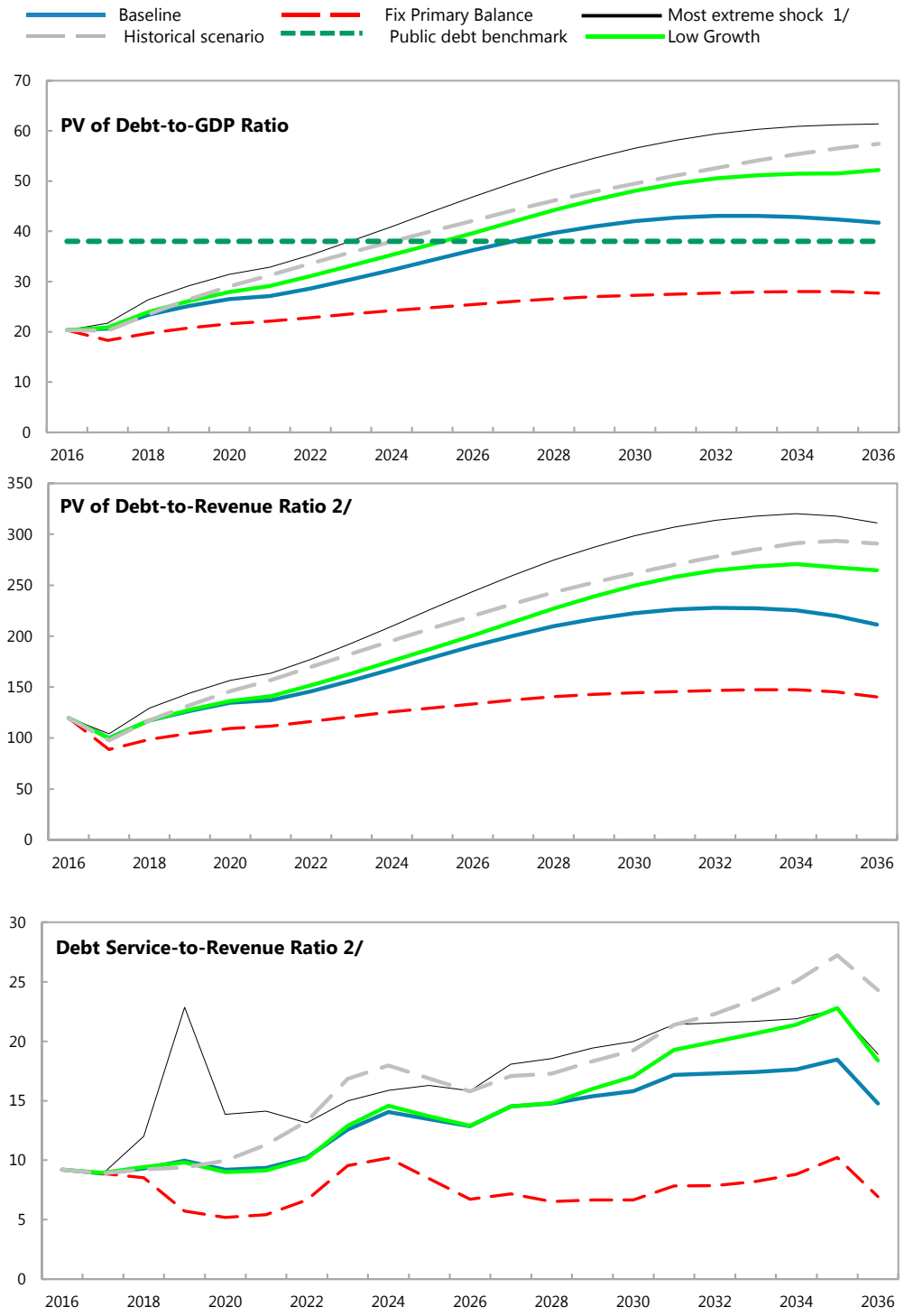
Figure 2. Haiti: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016-36 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a One-time depreciation shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock

Figure 3. Haiti: Indicators of Public Debt Under Alternative Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.